1. Identity

Document	STAR-PKG-STAR-POLYFIT-EN · v2.1 · 2026-11-01
Package	star-polyfit
Version	0.1.0
Capability	star.numerics.polyfit
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_polyfit

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_polyfit-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_polyfit-0.1.0-py3-none-any.whl	afad4259ffba242f9ff63d56e7ffb5712dc0ff0378ff5f5f0c56ab0fb6e92fff
star_polyfit.py	06546fdd6df439b32b38da7c47bacd00216407af317f25f0d0e1784605f0f566
star_polyfit-0.1.0-source.zip	ca1c3e4ae541b9790c60815bbc566bb062256a5aa250229933972f032270395e

5. What it does, and its limits

Least-squares polynomial of degree 0 to 10, weighted or not: the normal equations are formed and solved in exact rational arithmetic and each coefficient is rounded once, so conditioning costs no digit; exact evaluation and exact residual sum of squares; NIST Wampler1 reproduced exactly; every coefficient of 161 problems equal to SymPy's exact solution.

Exact coefficients are not a good model: a high-degree fit of noisy data oscillates, and the exact coefficients of a badly conditioned problem change a lot when the data change a little; the float nearest to each coefficient is still only a float, and evaluating the fitted polynomial in ordinary float arithmetic far from the origin can lose every digit (use `polyval`, or centre the data). No standard errors, no covariance matrix, no orthogonal polynomials, no automatic choice of the degree, no robust fitting. Exact arithmetic is slow: about a second for 2000 points at degree 10 on a laptop. "Nearest float" refers to the floats passed in, not to the decimal numbers you may have meant.

- `polyfit(xs, ys, degree, weights=None)` returns the coefficients, lowest power first, of the polynomial that minimises the sum of w (y - p(x))². `polyval(coefficients, x)` evaluates a polynomial; `residual_sum(xs, ys, coefficients, weights=None)` returns the weighted sum of squared residuals.
- The normal equations are formed and solved in exact rational arithmetic on the floats given, and each result is rounded once: the conditioning of the problem costs no digit. `polyval` and `residual_sum` are exact sums rounded once as well.
- Measured on 161 problems (120 well conditioned, 40 badly conditioned, and the NIST dataset Wampler1): every coefficient equals the exact rational solution of SymPy rounded to a float and the QR solution of mpmath at 120 digits. NumPy agrees within 1e-9 on the well-conditioned problems; on the badly conditioned ones (x in [1000, 1001], x = 1e6 + k) its coefficients share no digit with the exact ones (the numbers are in the evidence file).

- Every function returns floats or raises `ValueError`: a degree that is not an integer from 0 to 10, data that are not lists or tuples of finite real numbers within 1e100 (booleans refused), different lengths, negative weights, fewer distinct x with a positive weight than degree + 1, a result too large for a float.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_polyfit_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-polyfit}: Least-squares polynomial, correctly rounded},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_polyfit/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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